

One-Hot Encoding with Indian State Names (using Scikit-learn)

✓ 1 Install the Library

```
1 import pandas as pd
2 from sklearn.preprocessing import OneHotEncoder
```

✓ 2 Create a sample DataFrame with Indian state names

```
1 indian_states_data = {
2     'Customer_ID': [101, 102, 103, 104, 105, 106],
3     'State': ['Maharashtra', 'Delhi', 'Karnataka', 'Maharashtra', 'Uttar Pradesh', 'Tamil Nadu'],
4     'Purchase_Amount': [1500, 800, 2200, 1700, 1000, 3000]
5 }
6
7 df_indian_states = pd.DataFrame(indian_states_data)
8
```

✓ 3 Display the DataFrame

```
1 display(df_indian_states)
```

	Customer_ID	State	Purchase_Amount
0	101	Maharashtra	1500
1	102	Delhi	800
2	103	Karnataka	2200
3	104	Maharashtra	1700
4	105	Uttar Pradesh	1000
5	106	Tamil Nadu	3000

4 Initialize the OneHotEncoder

- `sparse_output=False` ensures a dense array output

```
1 state_encoder = OneHotEncoder(  
2     handle_unknown='ignore',  
3     sparse_output=False  
4 )
```

5 Fit and transform the 'State' column

- Reshape to 2D array as scikit-learn transformers expect

```
1 states_encoded = state_encoder.fit_transform(df_indian_states[['State']])
```

```
1 states_encoded
```

```
array([[0., 0., 1., 0., 0.],  
       [1., 0., 0., 0., 0.],  
       [0., 1., 0., 0., 0.],  
       [0., 0., 1., 0., 0.],  
       [0., 0., 0., 0., 1.],  
       [0., 0., 0., 1., 0.]])
```

6 Create a DataFrame from the encoded data

```
1 states_encoded_df = pd.DataFrame(  
2     states_encoded,  
3     columns=state_encoder.get_feature_names_out(['State'])  
4 )  
5
```

```
1 states_encoded_df
```

	State_Delhi	State_Karnataka	State_Maharashtra	State_Tamil Nadu	State_Uttar Pradesh
0	0.0	0.0	1.0	0.0	0.0
1	1.0	0.0	0.0	0.0	0.0
2	0.0	1.0	0.0	0.0	0.0
3	0.0	0.0	1.0	0.0	0.0
4	0.0	0.0	0.0	0.0	1.0
5	0.0	0.0	0.0	1.0	0.0

7 Concatenate with the original DataFrame (dropping the original 'State' column)

```

1 df_indian_states_encoded = pd.concat(
2     [df_indian_states.drop(columns=['State']),
3       states_encoded_df], axis=1
4 )
5

```

1 df_indian_states_encoded

	Customer_ID	Purchase_Amount	State_Delhi	State_Karnataka	State_Maharashtra	State_Tamil Nadu	State_Uttar Pradesh
0	101	1500	0.0	0.0	1.0	0.0	0.0
1	102	800	1.0	0.0	0.0	0.0	0.0
2	103	2200	0.0	1.0	0.0	0.0	0.0
3	104	1700	0.0	0.0	1.0	0.0	0.0
4	105	1000	0.0	0.0	0.0	0.0	1.0
5	106	3000	0.0	0.0	0.0	1.0	0.0

1 Start coding or [generate](#) with AI.

